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OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- Following the steps of data analysis to come up with this presentation.
- Methodology used:
 - Data Collection and Wrangling
 - Exploratory data analysis
 - Data Visualization
- Data visualization and dashboards supports this resulting presentation.
- Overall findings of the research will be discussed.
- Conclusion.

INTRODUCTION



- Technology is changing day by day. In order to keep pace with the ever changing tech, data has been gathered regularly.
- Through analysis of job postings, training portals, and surveys, several trends have been identified.
- Trend identification through data analysis can also predict the potential future of technology and what applications and languages will be used more in the future.
- This analysis can also discover what are trending programming languages, database applications, and IDE's.

METHODOLOGY



- Collecting data through web scraping and API's
- Data Wrangling
- Exploratory data analysis: finding correlations, analyzing data distributions, and outliers.
- Data visualization:
 - Individual analysis graphs using Python (matplotlib, seaborn)
 - Dashboards using IBM Cognos Analytics

RESULTS

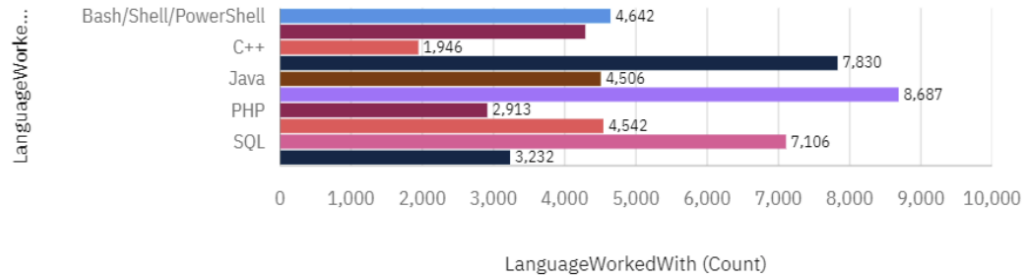


PROGRAMMING LANGUAGE TRENDS

Current Year

Top 10 Languages Worked With

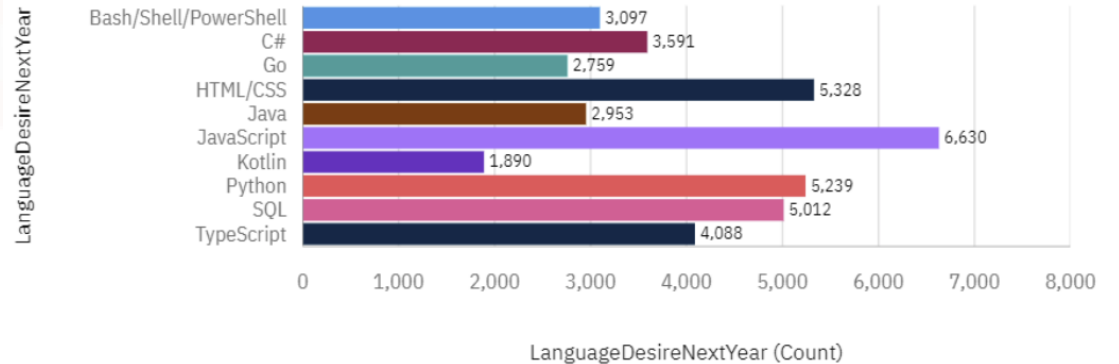
LanguageWorkedWith



Next Year

Top 10 Languages Desired To Learn

LanguageDesireNextYear



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- JavaScript and HTML/CSS tops the list of present popular programming languages, but there are trends on migrating to TypeScript.
- SQL is also a significant leader over Java/Python, present and future.
- C++ is losing grounds.

Implications

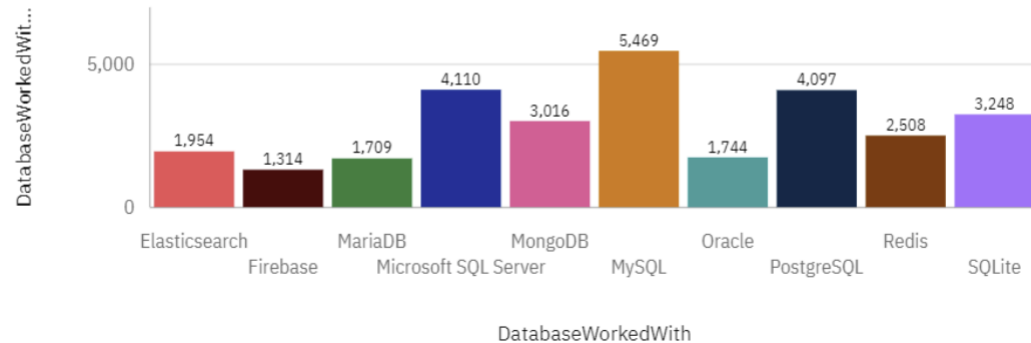
- JavaScript and HTML/CSS will continue to be popular.
- Interest in SQL, Python, and TypeScript are fast-growing.

DATABASE TRENDS

Current Year

Top 10 Databases Worked With

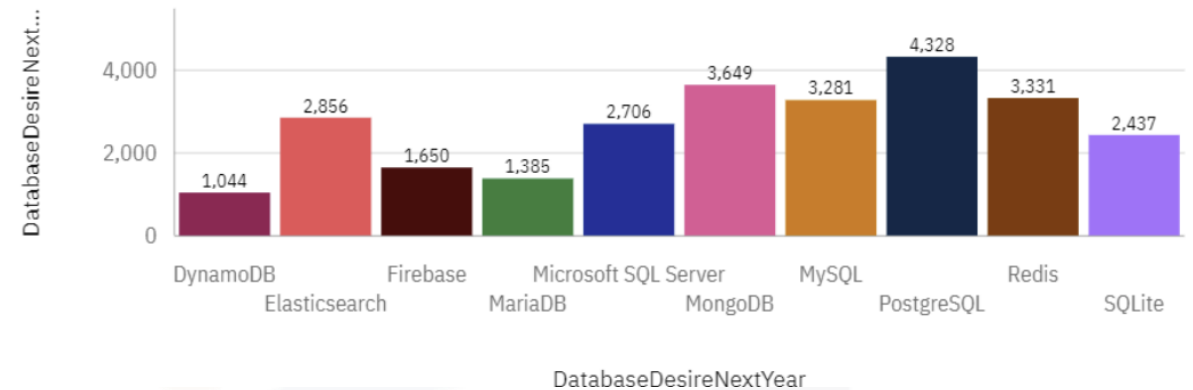
DatabaseWorkedWith



Next Year

Top 10 Databases Desired To Learn

DatabaseDesireNextYear



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- MySQL is the most popular database currently.
- Interest in PostgreSQL is rising up.
- Microsoft SQL server and SQLite are also learned by a lot of people, despite their current loss of grounds.

Implications

- PostgreSQL could take over MySQL in the future.
- Microsoft SQL Server and SQLite could face interest downfall.

DASHBOARD

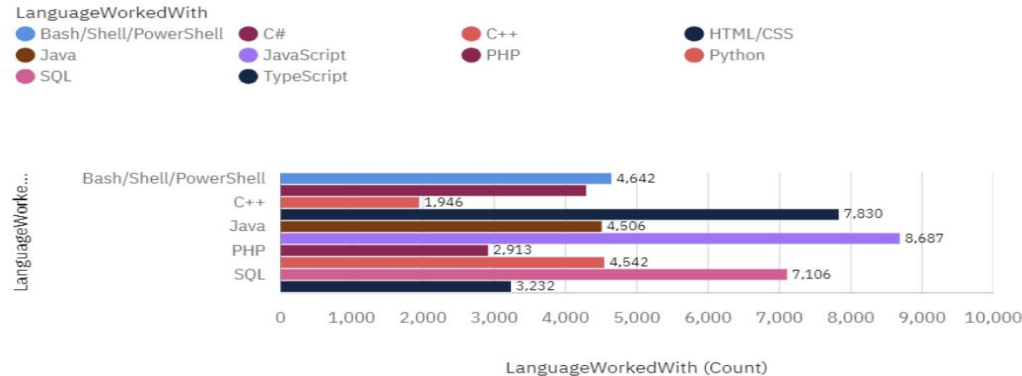


<https://github.com/spencerli688/IBM-Data-Analyst-Capstone-Project/blob/main/IBM%20Capstone%20Dashboard.pdf>

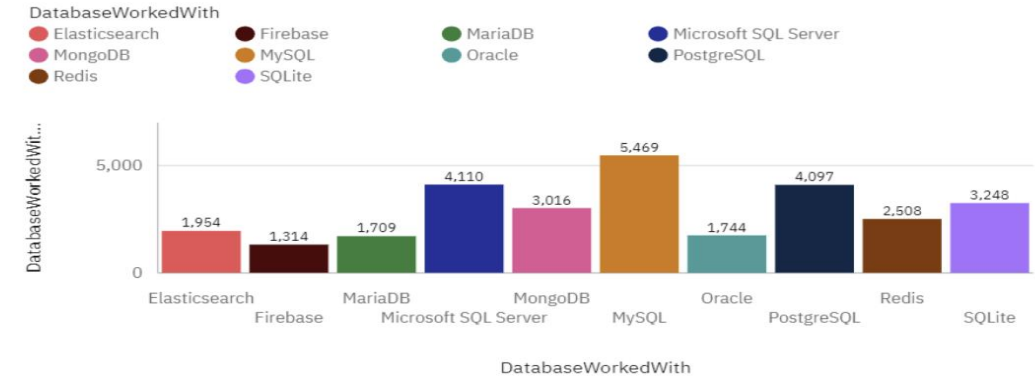
CURRENT TECH USAGE

Current Technology Usage

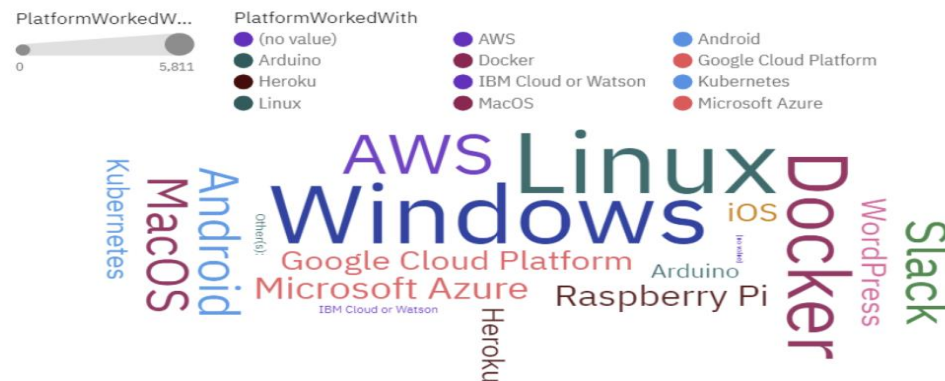
Top 10 Languages Worked With



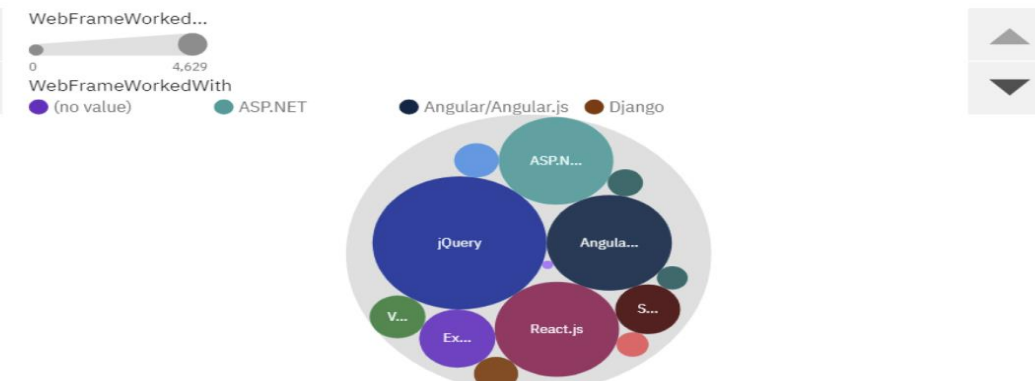
Top 10 Databases Worked With



Platform Popularity Word Cloud



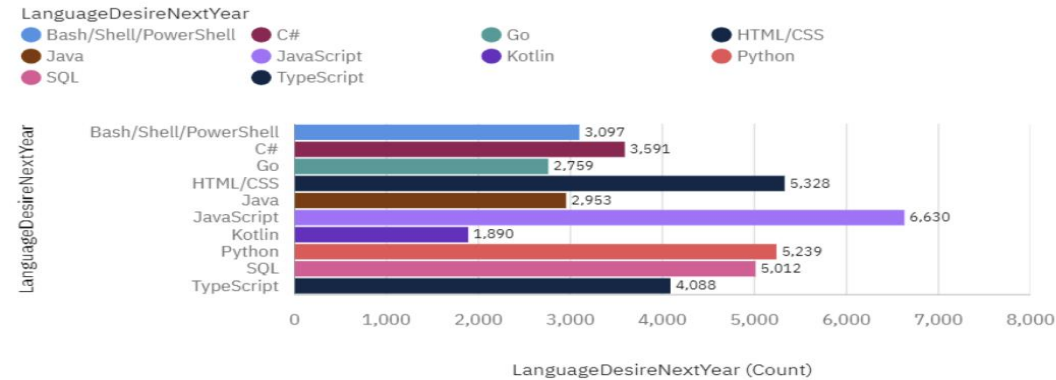
Top 10 Web Frames Worked With



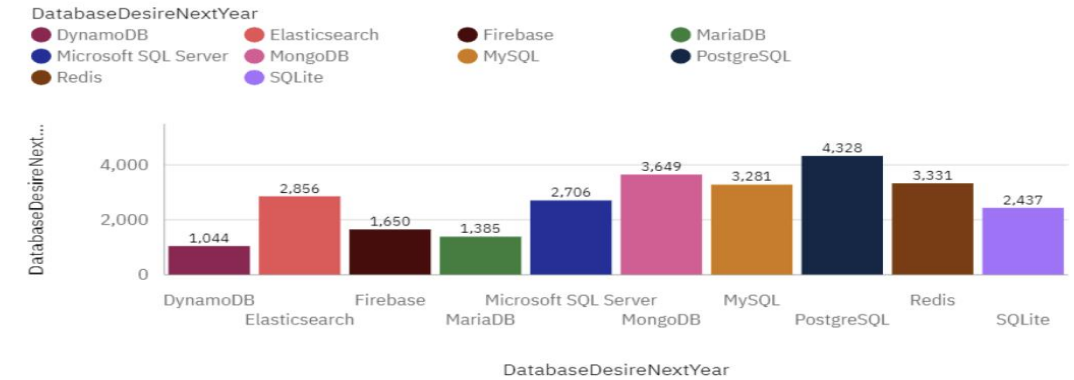
FUTURE TECH TRENDS

Future Technology Trends

Top 10 Languages Desired To Learn



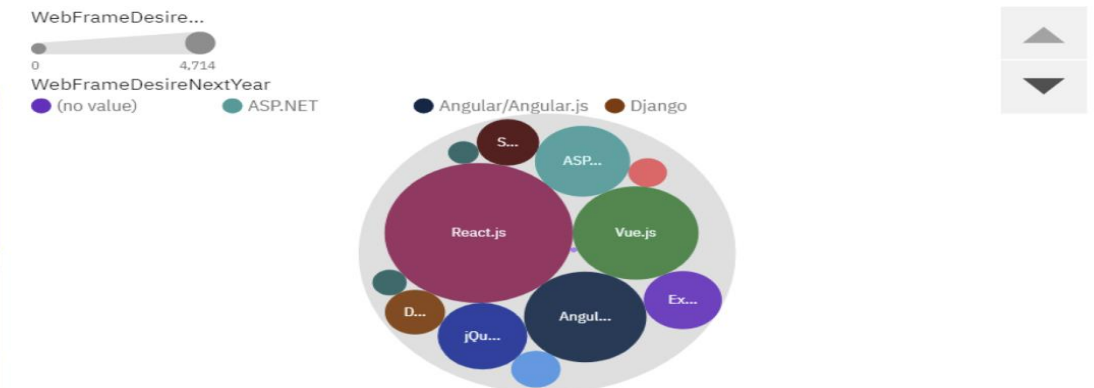
Top 10 Databases Desired To Learn



Platforms Desired To Learn On



Top 10 Web Frames Desired To Learn

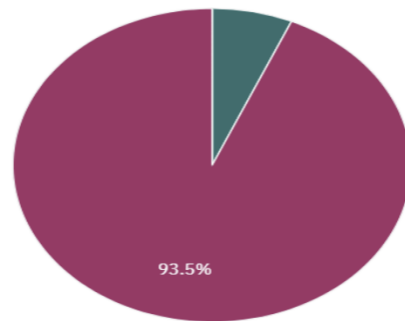


DEMOGRAPHICS DETAIL

Demographics

Respondent by Gender

Gender
● Woman ● Man

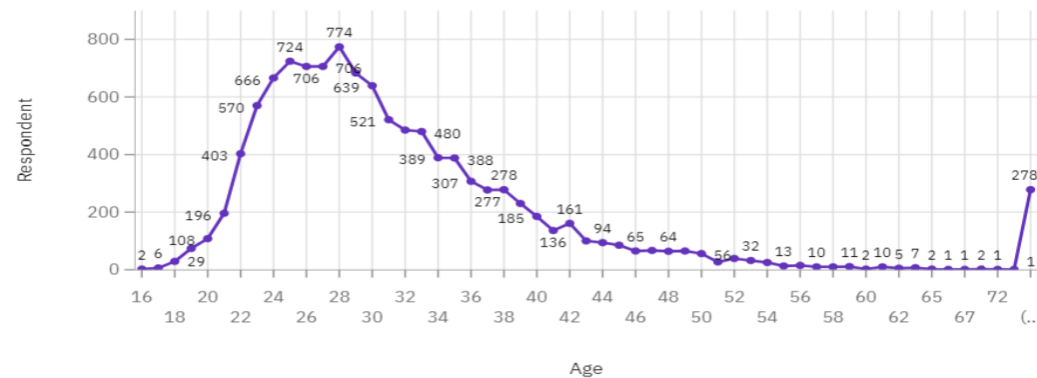


Respondent for Country regions

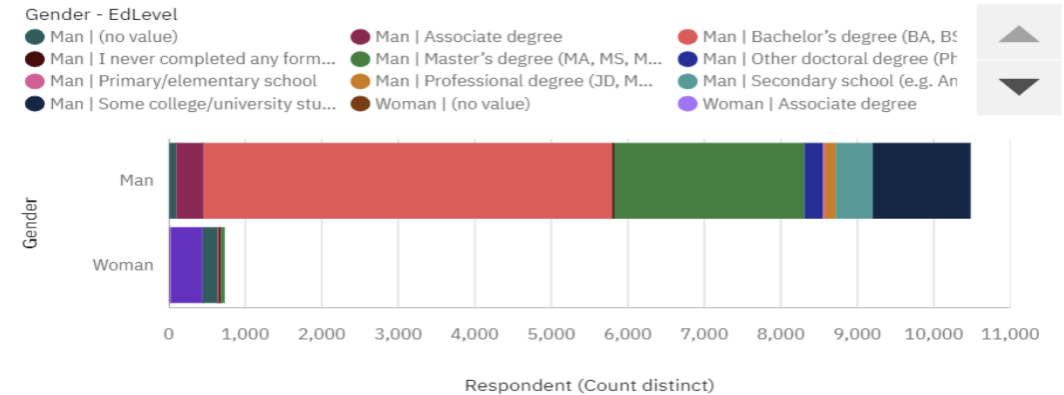
Respondent (Count)
1 3,058



Respondent by Age



Respondent by Gender colored by Gender and EdLevel



DISCUSSION



OVERALL FINDINGS & IMPLICATIONS

Findings

- JavaScript and MySQL are the two most used programming language and database (respectively) as of right now, but the latter's spot may be taken over by PostgreSQL if this trend keeps on going.
- Developers in the survey are mostly between 24 and 36, and are mostly male, and mostly from the United States of America.
- Most developers hold at least some postsecondary degree.

Implications

- Rising interest overall in TypeScript and PostgreSQL.
- Bachelors rising ground in the IT and developer community.
- Disproportionate gender and age composition may lead to some inequalities.

CONCLUSION

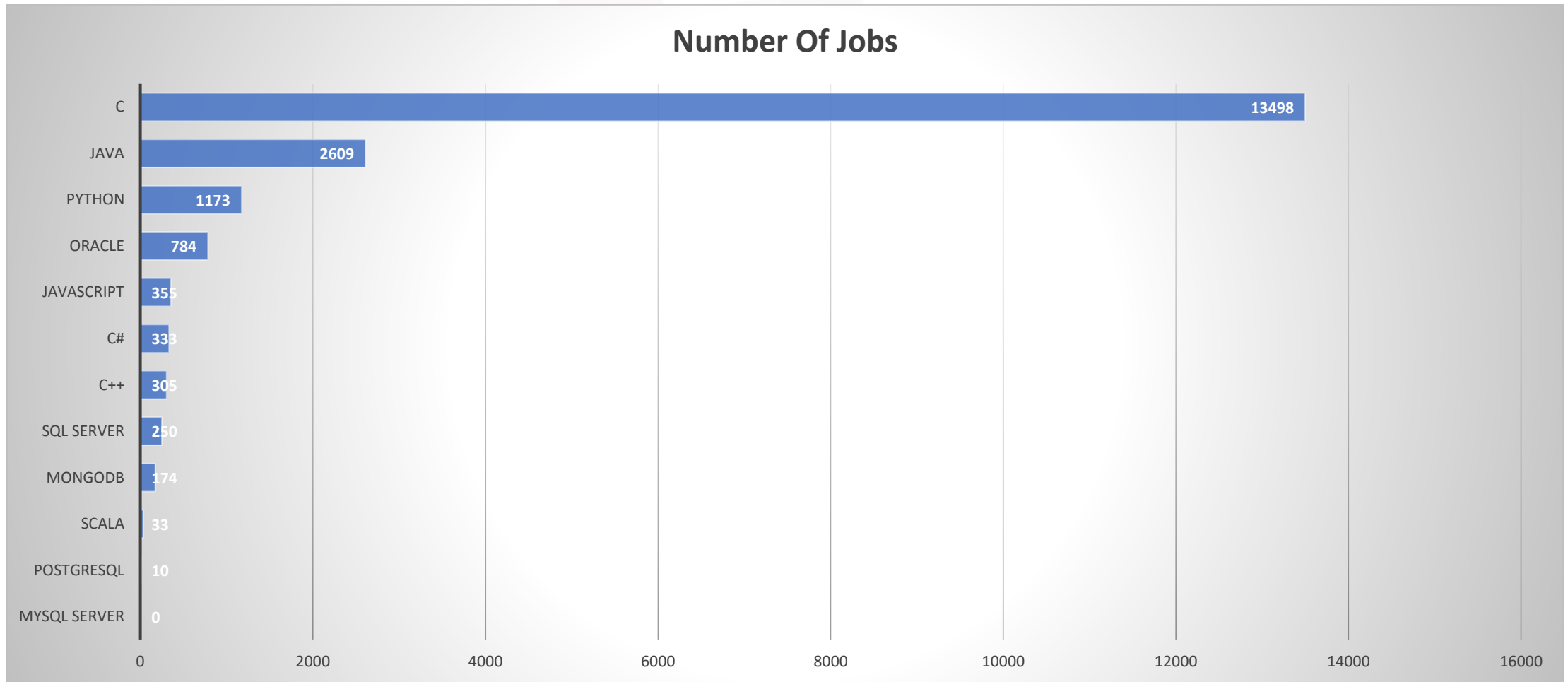


- Popularity trends of programming languages and databases can be identified with current and future data.
- Most developers have degrees, but more people with bachelors in related fields only are coming in.
- Work needs to be done in order for gender equality and accessibility in other countries to happen.

APPENDIX



JOB POSTINGS



POPULAR LANGUAGES

